Question 1:

Shenzhen Metro lines have many stations where you can transfer. Please output the `station_id` and `chinese_name` of each transfer stations between Metro Lines 1, 2, and 3, and also output those line pairs of `line_id`. Note that for connecting metro line pairs, the smaller sequence number comes first (`line_id1`), and the larger sequence number comes last (`line_id2`). Sort the returned results in ascending order of `station_id`. If `station_id` is the same, sort first by `line_id1` in ascending order, then by `line_id2` in ascending order.

深圳地铁线有很多站点可以换乘。请输出地铁123号线之间可以换乘的`station_id`和`chinese_name`，以及换乘的地铁线对的序号。注意换乘地铁线对中序号小的在前`line_id1`，序号大的在后`line_id2`。返回结果按`station_id`升序排序，`station_id`相同时先按`line_id1`升序排序，再按`line_id2`升序排序。

```
with a as (select distinct l1.station_id, l1.line_id as line_id1, l2.line_id as
line_id2
           from line_detail l1
                    join line_detail l2 on l1.station_id = l2.station_id
           where l1.line_id < l2.line_id and l1.line_id in (1,2,3) and
l2.line_id in (1,2,3))
select distinct stations.station_id, stations.chinese_name, a.line_id1,
a.line_id2
from a
         join stations on a.station_id = stations.station_id
order by stations.station_id, line_id1, line_id2;
```

station_id	chinese_name	line_id1	line_id2
3	老街	1	3
4	大剧院	1	2
9	购物公园	1	3
15	世界之窗	1	2
46	福田1	2	3

Question 2:

Find the average number of nearby bus stops for subway stations in each district, and calculate how many subway stations in each district have a number of nearby bus stops greater than the corresponding average value in each district. Subway stations with no nearby bus stops are not considered in this question.

The result set should include `district` and the count of subway stations with more than the average number of bus stops, sorted by `district`.

找到每个区的地铁站附近公交车站的数量的平均值。并计算每个区中有多少个地铁站附近公交站数量大于对应的平均值。这道题目不考虑附近没有公交站的地铁站。

返回结果集包含`district`,以及大于平均值的地铁站的数量，安装`district` 排序。

district	count
Bao'an	12
Futian	21
Longgang	11
Longhua	3
Luohu	9
Nanshan	12

```
select  b.district, count(*)
from (select a.station_id, bus_cnt, district, avg(bus_cnt) over (partition by 
district) as avg
      from (select station_id, count(*) bus_cnt
            from bus_lines bl
            group by bl.station_id) as a
               join stations s on s.station_id = a.station_id) as b
where bus_cnt > avg
group by b.district
order by district;
```

Question 3:

For each subway line, find the subway station with the smallest `station_id` and the subway station with the largest `station_id`. The result set contains `line_id`, the `english_name` of the subway stations with the smallest and largest `id`, and is sorted in ascending order by `line_id`.

在每一条地铁线上，找到`station_id`最小的地铁站与`station_id`最大的地铁站。结果集里包含`line_id`, id最小、最大的地铁站的`english_name`， 并按照`line_id`升序排列。

line_id	english_name	english_name
1	Luohu	Airport East
2	Grand Theater	Xinxiu
3	Laojie	Shuanglong
4	Convention and Exhibition Center Station	Niuhu
5	Qianhaiwan	Yijing
7	Chegongmiao	Wenti Park
9	Chegongmiao	Wenjin
11	Grand Theater	Shanghai Hotel

```sql
select ll.line_id, s1.english_name, s2.english_name
from (select distinct line_id,
                      min(l.station_id) over (partition by line_id) min,
                      max(l.station_id) over (partition by line_id) max
      from line_detail l) x
         join stations s1 on x.min = s1.station_id
         join stations s2 on x.max = s2.station_id
         join lines ll on ll.line_id = x.line_id
order by x.line_id;
```

Question 4:

Find the top three subway stations in Futian, Nanshan, and Luohu districts with the most bus stops near their respective subway stations (rank), and the count of their corresponding bus stop. Return a result set containing `district`, the subway station's `english_name`, and the number of bus stops, then sorted in ascending order by `station_id`.

找出福田、南山、罗湖区里地铁站附近公交站数量前三多(rank)的地铁站，及其公交站数量。返回结果集包含`district`, 地铁站`english_name`, 以及公交站的数量, 并按照`station_id` 升序排列。

```
select b.district, b.english_name, b.bus_cnt
from (select sub.station_id,
             s.english_name,
             sub.bus_cnt,
             s.district,
             rank() over (partition by district order by bus_cnt desc) as rnk
      from stations s
               join(select station_id, count(*) as bus_cnt
                    from bus_lines bl
                    group by bl.station_id) sub on s.station_id =
sub.station_id
      where s.district in ('Luohu', 'Nanshan', 'Futian')) as b
where rnk <= 3
order by station_id;
```

district	english_name	bus_cnt
Luohu	Luohu	44
Luohu	Guomao	48
Futian	Gangxia	39
Futian	Zhuzilin	59
Nanshan	Window of the World	55
Nanshan	Hi-Tech Park	52
Nanshan	Shenzhen University	49
Luohu	Caopu	54
Futian	Huangmugang	50

Question 5:

How many subway stations have **1, 2, 3, ..., n `connection`s** in their vicinity?

Based on the result set above, generate a sort number `row_number()` in **descending** order of the count of connections. Find the percentage increase between the number of subway stations with `row_number` of `n` and the number of subway stations with `row_number` of `n+1`. Round the percentage to one decimal place.

The returned result set contains the count of connections, their corresponding count of subway stations, the percentage increase, and the `row_number`.

附近`connection`数量分别是是1，2，3...n的地铁站分别有多少个?

在我们了解上述结果集的基础上，按照connection的数量降序生成一个排序号`row_number()`。找出row_number 是 `n`的地铁站的数量与row_number是`n+1`地铁站的数量的增幅百分比。百分比四舍五入保留1位小数。

返回结果集包含 connection数及其对应的地铁站数量，增幅百分比，与row_number。

```
select *, round(100.0 * (count - lag(count, 1) over ()) / count, 1) || '%' as
percentage
from (select row_number() over (order by connection_cnt desc) as row_number, *
      from (select connection_cnt, count(*) as count
            from (select s.station_id, count(*) connection_cnt
                  from stations s
                           join connections c on s.station_id = c.station_id
                  group by s.station_id) sub
            group by connection_cnt
            order by connection_cnt desc) as b) as c;
```

connection_cnt	count	percentage	Row_number
8	1	Null	1
7	1	0.0%	2
4	6	83.3%	3
3	3	-100.0%	4
2	23	87.0%	5
1	26	11.5%	6